



APPLICATION: KAIROS

AN-K001

KAIROS ALARM EVENTS SUBSYSTEM

Version 1v0

WARNING

The information in this document is subject to change without notice and its only purpose is to describe the item specified in the *Preliminary* section.

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This document has been written assuming that users have a basic knowledge of:

- ∞ analog and digital data transmission,
- ∞ data networks (LANs),
- ∞ IP-based networks,
- ∞ data exchange protocols and standards,
- ∞ basic concepts of electronic.

This document is not intended to deepen the above knowledge, unless some brief clarification where needed.

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PRELIMINARY

This document is intended as a brief guide to explain the use of **Alarm Events Subsystem** and to specify its **SNMP** and **ON-AIR** extensions.

This Subsystem is currently implemented on:

- ∞ the **Radio Activity RA-XXX** DMR/ANALOG Multiprotocol Base Stations (blue series),
- ∞ the **Radio Activity RA-TI** DMR Gateways,
- ∞ the **Radio Activity KAIROS** DMR/ANALOG Multiprotocol Base Stations.

This document will illustrate the implementation on the KAIROS Multiprotocol Base Stations.

For the implementation on the **RA-XXX** DMR/ANALOG Base Stations, please refer to the Application Note *AN_0002-RA-XXX_Alarm_Events_and_SNMP.pdf*.

For the implementation on the **RA-TI** DMR Gateways, please refer to the Application Note *AN_0011-RA-TI_Alarm_Events_and_SNMP.pdf*.

The **Alarm Events Subsystem** illustrated here is available since the release **1.6.0.0** of the **ka_bst** software, and can be managed since the release **0.9.164** of the **KAIROS_Manager** Remote Control Tool. The Alarm Events Subsystem shown below refers to those software versions released by **Radio Activity**.

All the text messages shown in this documents and manageable by **KAIROS_Manager** must be composed by **ANSI** characters (UNICODE is not supported).

The SNMP version used by **KAIROS** is **SNMPv1**.

ALARM EVENTS THEORY

The software on a **KAIROS** Base Station continuously monitors a lot of functional parameters, which can take the state of *good* or *bad* during a period. We can divide them into two major groups: *analog* parameters and *logic* parameters.

The *analog* parameters can take any value between a minimum and a maximum (i.e.: voltage, temperature...); the range of values is theoretically between $-\infty$ and $+\infty$, but in our case the range is limited to a minimum and a maximum detectable values, which are “reasonable” limits. An analog parameter has a range of normal values (that is a more restricted range than that of the detectable values), within which the parameter is considered *good*, and outside which it is considered *bad*; moreover, the parameter can have a hysteresis - related to its trend over time - that is a threshold value for the trend in one direction, outside which the parameter is *bad*, and a different threshold value for the trend in the opposite direction, inside which it is *good* again. Just for example, the internal temperature of the Base Station can be considered *bad* over 70 °C (158 °F) when increasing and *good* under 65 °C (149 °F) when decreasing.

The *logic* parameters normally point out specific facts occurred over time (i.e.: the activation of an input contact); they don't have a range of values: their state (*good* or *bad*) depends on whether they are active or not.

The state change of a parameter (from *good* to *bad* or vice versa) will produce an event, that can be a **RAISE Event**, if going from *good* to *bad*, or a **CLEAR Event**, if going from *bad* to *good*.

In addition, other parameters related to the Radio Network and DMR operations are monitored (i.e.: registration of a Slave BS, DMR Emergency Call...), and can produce events when detected. They are generally considered as **Notice Events** and are analogous to **Alarm RAISE** events; the main difference is that they don't produce a **CLEAR Event**.

The produced Alarm or Notice Events can be dispatched to the user in various ways, as shown later in the document.

SEVERITY DEGREE OF AN ALARM EVENT

Every event has its own default severity degree, which specifies the relevance of the event; the following table shows the possible severity degrees:

Severity	Value	Meaning
NOTICE	0	The lowest degree. It is used for significant events
WARNING	1	This degree specifies an event requiring attention. It must be watched
MINOR	2	This degree specifies an event indicating a fault for which a fixing is needed, but not urgent
MAJOR	3	This degree specifies an event indicating a fault for which a fixing is needed and urgent
CRITICAL	4	The highest degree. It specifies an event indicating a very serious fault: all the possible resources must be used to fix it quickly

SNMP EXTENSION

The SNMP Extension can simply send an **SNMP Trap** related to an Alarm or Notice Event to one or more SNMP Servers that have been set with a proper **MIB** file.

ON-AIR EXTENSION

Currently the **ON-AIR** extension can send to a radio terminal (portable or Mobile/Fixed) a **DMR Text Message** related to the Event.

ALARM EVENTS SETUP

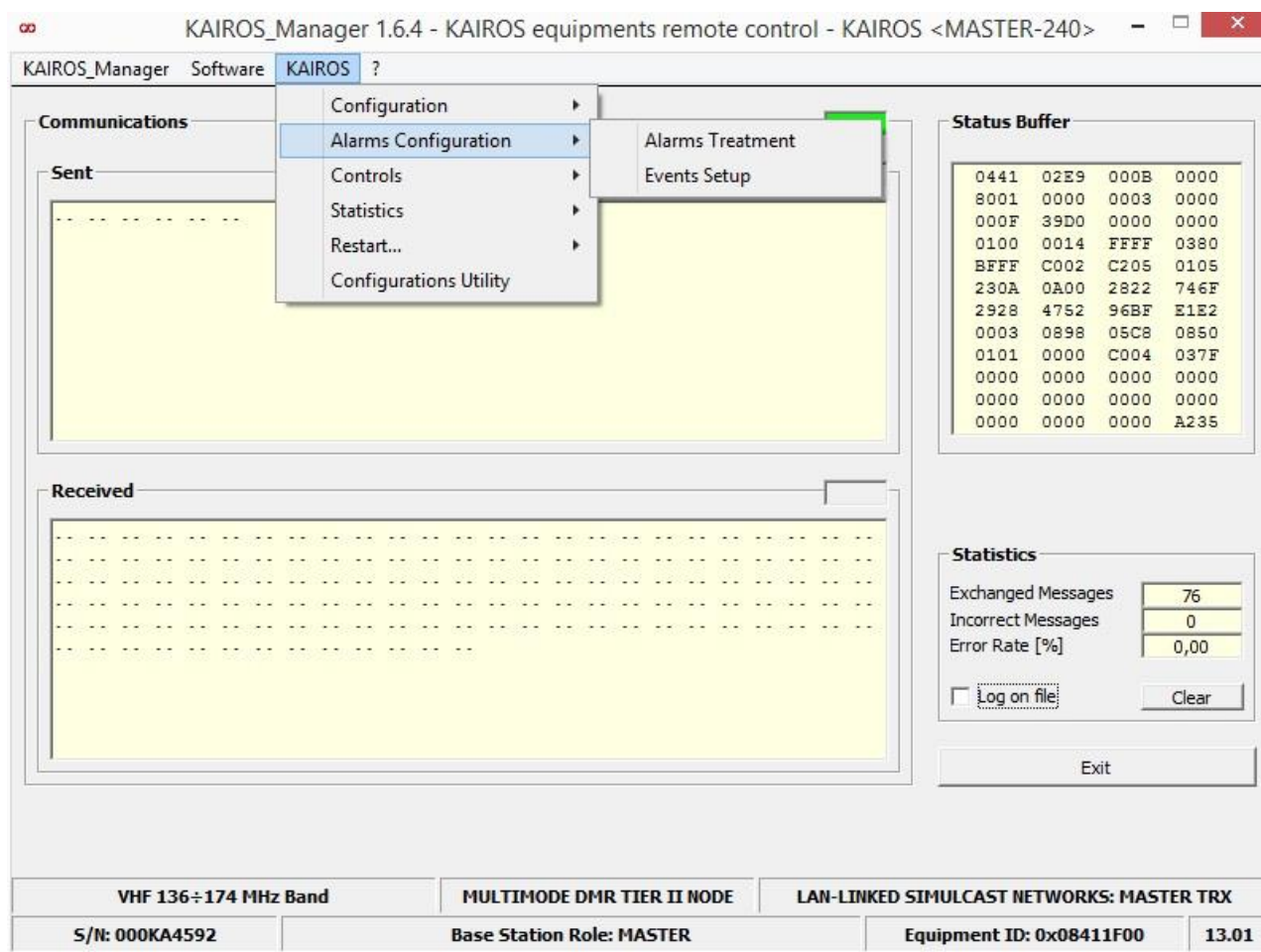
For every event, the user can:

- ∞ decide whether he should be notified of,
- ∞ define in which way the notification should happen,
- ∞ define some parameters related to the notification.

This setup can be done by the **KAIROS_Manager** Remote Control Tool connected to the Base Station.

ALARMS CONFIGURATION

From the Main window, click **KAIROS→Alarms Configuration**:



You have two further menu items:

Alarms Treatment: sets the general parameters of the Alarm or Notice Events notification,

Events Setup:..... configures how to handle and dispatch every single event.

ALARMS TREATMENT

From the Main window of **KAIROS_Manager**, click **KAIROS→Alarms Configuration→Alarms Treatment**; this window will appear:

As mentioned above, by this window you currently can set the dispatching mode of an Event, in terms of *SNMP Traps* and *DMR Text Messages*, as shown in the following tables.

<SNMP Traps / Community Data> Section

Item	Description	Default
Name	Name of the SNMP Community	"public"
Password	Password of the SNMP System	"admin"

<SNMP Traps / SNMP Servers> Section

You can define up to **8** SNMP Servers which an SNMP Trap would be sent to, by both IP Address and IP Port number:

Item	Description	Default
Present	If you want to add a server, specifying its IP Address, you at first must check this checkbox; if you want to remove a server, simply uncheck this checkbox	Unchecked
IP Address	Specifies the IP Address of an SNMP Server to which the Trap will be sent	--
IP Port	Specifies the IP Port number (in the range 20÷65000) of the server to which the Trap will be sent	162

<SNMP Traps / SNMP Parameters> Section

Item	Description	Default
Threshold level	Specifies the severity level under which an SNMP trap will not be sent	NOTICE
Common Severity for CLEAR	Specifies a unique severity degree for all the <i>CLEAR Events</i> , which takes priority over the per-Alarm <i>CLEAR</i> severity degree (in addition to the already shown severity degrees (see above, Severity Degree of an Alarm Event), this list-box provides an <i>UNDEFINED</i> value, which means that a <i>CLEAR Event</i> will assume the severity degree specified in the <i>Events Setup</i> window).	UNDEFINED

<DMR Messages / Destination> Section

You can define up to 4 DMR Destination which a DMR text message will be sent to, by ID, Timeslot, Type and Format:

Item	Description	Default
Present	If you want to add a DMR Destination, you first must check this checkbox; if you want to remove a DMR Destination, simply uncheck this checkbox	Unchecked
Destination ID	Specifies the Destination ID, which can be an Individual ID or a Group ID, according to the Destination type	--
Timeslot	Specifies the Timeslot (A or B) which a text message will be sent on	--
Destination Type	Specifies the type of the Destination ID (currently the only choice is between "ALL CALL" and "GROUP ID")	
Message Format	Specifies the format of the text messages that will be sent (currently the only choice is between MotoTRBO™ format and ETSI format)	162

<DMR Messages / Message Parameters> Section

You can define some additional parameters for an event dispatched by a DMR text message:

Item	Description	Default
Prepend Date	If checked, the related text of an Event will be prepended by the UTC date of an Event occurrence, in ISO 8601 format (YYYY-MM-DD)	Checked
Prepend Date	If checked, the related text of an Event will be prepended by the UTC time of an Event occurrence, in ISO 8601 format (HH:MM)	Checked
Base Station Identification	You can choose here how the Base Station will be identified in the DMR text message sent; the current choices are: None no identification string will be inserted Equipment Name the name of the Station, as set in the Main Setup window Station ID the Station ID value, as set in the Main Setup window Equipment S/N the Serial Number of the Station, assigned from Factory, normally displayed in the main window of KAIROS_Manager Equipment ID (hex) the Equipment ID of the Station, assigned from Factory, normally displayed in hexadecimal format in the main window of KAIROS_Manager Custom String the identification will be the string chosen by the user, set in the following item	Equipment Name
Custom ID	Sets your own Identification String to be sent if Custom String has been chosen in the previous item	"KAIROS"

EVENTS SETUP

From the Main window of **KAIROS_Manager**, click **KAIROS→Alarms Configuration→Events Setup**; this window will appear:

Events Setup - KAIROS <MASTER-240>

Supply Overvoltage | Supply Undervoltage | **Ethernet Link Status** | PLD Data Correctness | DSP Loading

SNMP Traps

Traps Destination

172.33.2.105:162	☐
172.33.2.44:162	☒
	☐
	☐
	☐
	☐

Traps Sending Policy

Merge Auxiliary Data in Text Messages ☒

Send the CLEAR TRAP on a CLEAR Event ☒

Repeat RAISE TRAP every

On-Air Messages

DMR

Destination	TS	Call Type	Format
101	A	Group	MotoTRBO
16777215	A	All Call	ETSI

Repetitions

Interval [s]

Properties and Methods

Alarm Enabled ☒

Consecutive events to toggle

Insert into ACTIVE/HOT-SPARE analysis ☐

Alarm Raise

Message

Broken Ethernet Link

System Defined Severity

User-Defined Level

Alarm Clear

Message

Ethernet Link Restored

User Defined Severity

User-Defined Level

I/O Contacts Activation

Activate EXT-contact ☒

Write Read Close

In this window you can decide how to dispatch an Alarm or Notice. The Base Station can send an alarm:

- ∞ through LAN, to one or more configured SNMP Servers,
- ∞ through a DMR RF channel to a single terminal or a group of terminals,

Below you can find the meaning of all the items.

<SNMP Traps / Traps Destination> Section

In this section you will find all the previously configured SNMP Servers (IP Address + IP Port number): if you want the Station to send the trap related to the Event to one or more of the listed servers, simply check the checkboxes associated to the servers. If you don't want the Station to send the trap, then uncheck all.

<SNMP Traps / Traps Sending Policy> Section

For every Alarm Event, in this section you can decide:

- ∞ if the additional data (if present) sent by the dedicated **OID**, is also added to the event-related text message (for additional data entity, see below, **Additional Data and Consecutive Events to Toggle** and **Additional Data Fields**),
- ∞ if the Subsystem has to repeat the **RAISE** TRAP at a specified interval until the generating event is present, or it has to send the **RAISE** TRAP only when the alarm condition occurs and send the **CLEAR** TRAP once the condition ends. This is only possible for Alarms where a CLEAR Event is present

<On-Air Messages / DMR> Section

In this section you will find all the previously configured DMR Destinations (ID + Timeslot + Type + Format): if you want the Station sends to one or more of the listed destinations the event-related text message, simply check the checkboxes associated to the destinations. If you don't want the Station sends the message, uncheck all.

In addition, there are two further parameters, specific for each Event:

<i>Item</i>	<i>Description</i>	<i>Default</i>
Repetitions	Specifies how many times the message will be sent, in the range 1÷4	1
Interval [s]	Specifies the time in seconds between the repeated sending of the message, in the range 3÷48, in increments of 3; it is meaningful only if the <i>Repetitions</i> item is set to more than 1	--

<Properties and Methods> Section

In this section you will define the behavior of an Event, as follows:

<i>Item</i>	<i>Description</i>	<i>Default</i>
Alarm Enabled	Enables or disables all the actions subsequent to the detection of the Event	Checked
Consecutive Events to Toggle	Specifies how many consecutive detections of the Event are needed to toggle the status of the Alarm; note that in some Events this parameter is not enabled to the User (see Additional Data and Consecutive Events to Toggle)	(Distinct for each Event)
Insert into ACTIVE/HOT-SPARE Analysis	Here you can specify if the Event participates to the Equipment Status Analysis which can produce an ACTIVE-to-SPARE role exchange in a 1+1 environment; please, note that not all Events are qualified to participate to that analysis, and currently not all qualified Events, even selectable, will participate to the analysis: it is limited to the followings: ∞ Ethernet Link Status ∞ No TX Power ∞ TX Power too low ∞ TX Power too high ∞ TX SWR Alarm ∞ TX Power Reduction ∞ BS Temperature ∞ TX Temperature	Unchecked

<Alarm Raise> and <Alarm Clear> Sections

<i>Item</i>	<i>Description</i>	<i>Default</i>
Message	Here you can specify the text of the messages that will be sent to configured SNMP Server(s) and through the DMR RF channel; you can insert a free-format ANSI text up to 47 alphanumeric characters. Please note that some Alarms don't have a CLEAR event: in this case an appropriate reminder will be displayed instead of the <i>Message</i> textbox of the <i>Alarm Clear</i> section	(Internal string)
User-Defined Severity	Here you can specify the severity degree for the displayed alarm. While the user can define the CLEAR Event severity degree for every alarm (optionally overwritten by the <i>Common Severity for CLEAR</i> value, see above, <SNMP Traps / SNMP Parameters> Section ; in this case the associated label changes its value from " <i>User Defined Severity</i> " to " <i>Common Severity Degree</i> "), the user can define the RAISE Event severity degree only for some alarms, because for some others it is still Factory-defined and cannot be changed (in this case the degree is anyway displayed, but the up-down roller is disabled and the associated label changes its value from " <i>User Defined Severity</i> " to " <i>System Defined Severity</i> "). The range of values is as shown above, in Severity Degree of an Alarm Event . If you didn't specify any SNMP Server in the <i>Alarms Treatment</i> window, the <i>User-Defined Severity</i> value has no meaning	Factory Assigned (see below, Handling the Severity Degree)
User-Defined Level	Here You can specify a value, in the range 0÷15, that will be inserted into the SNMP Trap; the SNMP Server can use this value as an additional filter, or a customized priority/significance for that alarm. If you didn't specify any SNMP Server in the <i>Alarms Treatment</i> window, the <i>User-Defined Level</i> value has no meaning	0

<I/O Contacts Activation> Section

<i>Item</i>	<i>Description</i>	<i>Default</i>
Activate EXT Contact	If checked, the <i>RAISE Event</i> of the alarm will participate in the activation of the ALARM OUT contact on the rear DB25 connector (this is ORed with all the alarms for which this box is checked); the <i>CLEAR Event</i> of the alarm will release the activation of the contact (this is ANDed with all the alarms for which this box is checked).	Unchecked

ALARM EVENTS SPECIFICATIONS

Let's have a first look to the handled Events that can produce Alarms or Notice.

The following table shows a summary of all the alarm events handled by the current **ka_bst** software; to correctly read it, please look at this legend:

Significant for (Roles):

ANY: ...independent from the Role

MS: MASTER Base Station

SM: SUBMASTER Base Station

SL: SLAVE Base Station

LD: LINK-DOWN Base Station (Master Side Link Station)

LU: LINK-UP Base Station (Satellite Side Link Station)

BU: SLAVE Base Station defined as *BACKUP MASTER* when it is promoted to *MASTER*

MF: MOBILE/FIXED Base Station

A/N (Alarm/Notice):

A: Alarm

N: Notice Only (the **CLEAR Event** is **not** provided for this Event)

Here is the table:

Event Number	Event Label	Significant for (Roles)	Description	A/N
1	I/O 1 Contact	ANY	It rises on the closure of rear Input Contact 1 and clears on its opening	A
2	I/O 2 Contact	ANY	It rises on the closure of rear Input Contact 2 and clears on its opening	A
3	Logic Supply Status	ANY	It rises when the internal voltage regulator for logic devices supply detects a failure and clears otherwise	A
4	Supply Undervoltage	ANY	It rises if the main supply voltage decreases under 10.8 V or the logic supply voltage decreases under 2.8 V, and clears when the main supply voltage increases over 11.1 V and the logic devices supply voltage increases over 2.9 V	A
5	Supply Overvoltage	ANY	It rises if the supply voltage increases over 15.6 V or the logics supply voltage increases over 3.5 V, and clears when the main supply voltage decreases under 15.3 V and the logics devices supply voltage decreases under 3.4 V	A
6	Ethernet Link Status	ANY	It rises when the Ethernet link looks disappeared or broken and clears when the Ethernet link is present. Of course it is meaningful only if the Base Station is connected through LAN to a network device	A
7	PLD Data Correctness	ANY	It rises when the PLD detect some incorrect programming data and clears otherwise	A
8	DSP Loading/Starting	ANY	It rises in case of problems to load the DSP Program or on failure on DSP startup and clears otherwise	A
9	GNSS Status	ANY	It rises on GNSS device failure and clears otherwise. It is significant only if the option board with the GNSS device is installed	A
10	Vocoders Status	ANY	It rises on Vocoders device failure and clears otherwise. It is significant only if the option board with the Vocoder devices is installed	A
11	BS Temperature	ANY	It rises when the temperature inside the Base Station, near the VCTCXO, will increases over 70 °C (158 °F) and clears when the temperature decreases under 65 °C (149 °F). This Event will produce an automatic progressive power reduction, until the temperature will decrease under the limit	A
12	TX Temperature	ANY	It rises when the temperature of the power stage of the transmitter increases over 85 °C (185 °F) and clears when the temperature decreases under 80 °C (176 °F). This Event will produce an automatic progressive power reduction, until the temperature will decrease under the limit	A
13	No TX Power	ANY	It rises if the transmitter is engaged (a transmission is requested), but no output power is emitted, and clears otherwise. The detection occurs only when the transmitter is engaged: this means that the status is retained when no transmission is requested	A

Event Number	Event Label	Significant for (Roles)	Description	A/N
14	TX Power too low	ANY	It rises if the transmitter is engaged (a transmission is requested), but output power is excessively lower than expected, and clears otherwise. The output power is checked against a lower limit related to the desired value. As a rule of thumb, consider the output power good if it is higher than about 50% of the desired value for values under 10 W and higher than about 75% otherwise. The detection occurs only when the transmitter is engaged: this means that the status is retained when no transmission is requested	A
15	TX Power too high	ANY	It rises if the transmitter is engaged (a transmission is requested), but output power is excessively higher than expected, and clears otherwise. The output power is checked against a higher limit related to the desired value. As a rule of thumb, consider the output power good if it is lower than about 150% of the desired value for values under 10 W and lower than about 120% otherwise. The detection occurs only when the transmitter is engaged: this means that the status is retained when no transmission is requested	A
16	TX SWR Warning	ANY	It rises if the transmitter is engaged (a transmission is requested) and the measured SWR is over 2.5 , and clears otherwise. The detection occurs only when the transmitter is engaged at a power higher than 1 W: this means that the status is retained when no transmission is requested	A
17	TX SWR Alarm	ANY	It rises if the transmitter is engaged (a transmission is requested) and the measured SWR is over 4.0 , and clears otherwise. The detection occurs only when the transmitter is engaged at a power higher than 1 W: this means that the status is retained when no transmission is requested. This Event will produce an automatic fixed power reduction to 1 W. Of course, if the desired output power is higher than 1 W, the check will be done at the desired power value	A
18	TX Power Reduction	ANY	It rises when the automatic power reduction is in effect and clears otherwise	A
19	Synchronization Source	ANY	It rises when the Base Station is locked on an alternate synchronization source (the current source is different from the primary source previously set) and clears otherwise	A
20	Synchronization Status	ANY	It rises when the Base Station is not locked on any of the possible sources previously set and clears otherwise	A
21	Board Vtunes Status	ANY	It rises when the DSP reports that a Vtune of any of the PLLs used for logic devices is out of range or can't handle it correctly, and clears otherwise	A
22	TRX Vtunes Status	ANY	It rises when the DSP reports that a Vtune of any of the PLLs used for RF devices is out of range or can't handle it correctly, and clears otherwise	A
23	Board Clocks Status	ANY	It rises when the DSP reports a failure on any of the clock generators used for logic devices and clears otherwise	A
24	TRX Clocks Status	ANY	It rises when the DSP reports a failure on any of the clock generators used for RF devices and clears otherwise	A
25	Board PLLs Lock Status	ANY	It rises when the DSP reports a <i>PLL unlock</i> on any of the clock generators used for logic devices and clears otherwise	A
26	TRX PLLs Lock Status	ANY	It rises when the DSP reports a <i>PLL unlock</i> on any of the clock generators used for RF devices and clears otherwise	A
27	PLD Status	ANY	It rises when the DSP reports a failure on the PLD device and clears otherwise	A
28	PLD<->DSP Chan Status	ANY	It rises when the DSP reports a failure on the communication channel between DSP and PLD, and clears otherwise	A
29	RX IFs Status	ANY	It rises when the DSP reports a failure on the IF modules of the PLD (for both Main or Diversity RX) and clears otherwise	A
30	RXs Status	ANY	It rises when the DSP reports a failure on the Main or Diversity RX and clears otherwise	A
31	RF Channel Noise	ANY	Not yet implemented	--
32	Registration to Master	SL,SM,LD	It rises when a Slave Base Station loose the logical LAN connection to its Master Base Station, and clears when it is re-connected	A
33	Registration of Slave	MS,SM,LU,BU	It occurs when a Slave Base Station connected through LAN has completed the <i>registration</i> process, that is the Slave will participate to the Radio Network	N
34	Unregistration of Slave	MS,SM,LU,BU	It occurs when a Slave Base Station connected through LAN has completed the <i>unregistration</i> process, that is the Slave will no longer participate to the Radio Network	N
35	Loss of Slave	MS,SM,LU,BU	It rises when a registered Slave Base Station, for any reason, no longer sends its <i>keepalive</i> message, so it looks disappeared	N
36	Master Role	BU	It rises when a Slave Base Station set as Backup Master becomes a Master and clears when it goes back to Slave	A
37	Connection to BK MST	SL,SM,LD	It rises when a Slave Base Station which has a defined Backup Master, once lost the logical LAN connection to its Master, connects to its Backup Master, and clears when it re-connects to its real Master	A

Event Number	Event Label	Significant for (Roles)	Description	A/N
38	DMR Emergency Call	MS,BU,MF	It occurs when the Base Station detects a DMR Emergency Call from a terminal on any Timeslot (A or B)	N
39	1+1 ACTIVE State	ANY	This Event is significant only if a couple of BSs are configured and linked one each other as 1+1 ACTIVE/HOT-SPARE BS, where one is normally active and the other is idle. It occurs when the BS enters into the ACTIVE state, that is, it is fully running	N
40	1+1 HOT-SPARE State	ANY	This Event is significant only if a couple of BSs are configured and linked one each other as 1+1 ACTIVE/HOT-SPARE BS, where one is normally active and the other is idle. It occurs when the BS enters into the HOT-SPARE state, that is, it is idle	N
41	TRX Layer Status	ANY	It rises when, for any reason (automatically or manually), the TRX software layer is disabled and clears when it is re-enabled. The reason will be inserted into the Additional Data	A
42	BS Layer Status	ANY	It rises when, for any reason (automatically or manually), the BS software layer is disabled and clears when it is re-enabled. The reason will be inserted into the Additional Data	A
43	SIP name Resolution	MS,MF	It rises if the Base Station can't resolve the Node Name of the SIP Server (can't get the IP Address from the name) and clears on a successful name resolution. It is significant only when the SIP Layer is active. The most common cause is a wrong or undefined IP Address of the Nameserver(s).	A
44	Fail to Register to SIP	MS,MF	It rises when the SIP Client into the Base Station, for any reason, can't register to the defined SIP Server. It is significant only when the SIP Layer is active	N
45	Registration to SIP	MS,MF	It rises when the SIP Client into the Base Station completed the registration process to the defined SIP Server. It is significant only when the SIP Layer is active	N
46	Unregistration from SIP	MS,MF	It rises when the SIP Client into the Base Station completed the unregistration process from the defined SIP Server. It is significant only when the SIP Layer is active	N
47	SIP Server Change	MS,MF	It rises if the Base Station point to the SIP Server by its Node Name and detects that the IP Address has changed. It is significant only when the SIP Layer is active	N
48	SIP TRUNK Server	MS,MF	It rises when the SIP Client into the Base Station can't access the SIP Trunk Server and clears otherwise. It is significant only when the SIP Layer is active and the TRUNK mode is selected	A

ADDITIONAL DATA AND CONSECUTIVE EVENTS TO TOGGLE

The **Additional Data** fields generally represent a measurement or informational value in two different ways (textual and numeric). For example, if the Event is related to an analog parameter, the content of the textual field is a literal string, eventually surrounded by parentheses, representing the value followed by the measurement unit (i.e.: **"(12.6 V)"**), while the content of the numeric field is a 32-bits signed integer (i.e.: **12600**), formatted according to these measurement units:

- ∞ Voltage in **mV**,
- ∞ Current in **mA**,
- ∞ Temperature in **°C/10** (tenths of degrees centigrade).

The **Consecutive Events to Toggle** amount specifies how many events must be detected consecutively for an Alarm before changing status, that is, produce a **RAISE** or **CLEAR** Event. It is used to minimize any false alarm and, while for some Alarms the amount is internally assigned and cannot be changed, for some other Alarms the User can assign a desired value.

The following table shows which Alarm Events the **Additional Data** fields are applied to, the Factory assigned default value for **Consecutive Events to Toggle** and the User capability for it:

<i>Alarm Number</i>	<i>Alarm Label</i>	<i>Additional Data Present</i>	<i>Additional Data Type</i>	<i>Example (numeric)</i>	<i>Example (textual)</i>	<i>Default Consecutive Events to Toggle</i>	<i>User can set a desired amount for Consecutive Events</i>	<i>Note</i>
1	I/O 1 Contact	NO	--	--	--	2	YES	
2	I/O 2 Contact	NO	--	--	--	2	YES	
3	Logic Supply Status	NO	--	--	--	2	NO	
4	Supply Undervoltage	NO	--	--	--	3	NO	1
5	Supply Overvoltage	NO	--	--	--	3	NO	1
6	Ethernet Link Status	NO	--	--	--	2	NO	
7	PLD Data Correctness	NO	--	--	--	1	NO	
8	DSP Loading/Starting	NO	--	--	--	1	NO	
9	GNSS Status	NO	--	--	--	1	NO	
10	Vocoders Status	NO	--	--	--	1	NO	
11	BS Temperature	NO	--	--	--	2	NO	2
12	TX Temperature	NO	--	--	--	2	NO	2
13	No TX Power	NO	--	--	--	3	NO	
14	TX Power too low	NO	--	--	--	3	NO	2
15	TX Power too high	NO	--	--	--	3	NO	2
16	TX SWR Warning	NO	--	--	--	3	NO	2
17	TX SWR Alarm	NO	--	--	--	3	NO	2
18	TX Power Reduction	NO	--	--	--	1	NO	
19	Synchronization Source	NO	--	--	--	2	YES	
20	Synchronization Status	NO	--	--	--	3	NO	
21	Board Vtunes Status	NO	--	--	--	2	NO	3
22	TRX Vtunes Status	NO	--	--	--	2	NO	3
23	Board Clocks Status	NO	--	--	--	2	NO	
24	TRX Clocks Status	NO	--	--	--	2	NO	
25	Board PLLs Lock Status	NO	--	--	--	3	NO	4
26	TRX PLLs Lock Status	NO	--	--	--	4	NO	4
27	PLD Status	NO	--	--	--	2	NO	
28	PLD<->DSP Chan Status	NO	--	--	--	2	NO	
29	RX IFs Status	NO	--	--	--	2	NO	
30	RXs Status	NO	--	--	--	2	NO	
31	RF Channel Noise	NO	--	--	--	2	YES	5
32	Registration to Master	NO	--	--	--	1	YES	
33	Registration of Slave	YES	UID of the Slave BS (num) Name of the Slave BS (txt)	0x08411f10	"SLAVE-241"	1	YES	
34	Unregistration of Slave	YES	UID of the Slave BS (num) Name of the Slave BS (txt)	0x08411f10	"SLAVE-241"	1	YES	
35	Loss of Slave	YES	UID of the Slave BS (num) Name of the Slave BS (txt)	0x08411f10	"SLAVE-241"	1	YES	
36	Master Role	NO	--	--	--	1	YES	
37	Connection to BK MST	NO	--	--	--	1	YES	
38	DMR Emergency Call	YES	DMR ID of the Sender	21	"ID 21"	1	YES	
39	1+1 ACTIVE State	NO	--	--	-	1	YES	
40	1+1 HOT-SPARE State	NO	--	--	--	1	YES	
41	TRX Layer Status	YES	Reason of enable/disable	4	"(no active channels)"	1	YES	
42	BS Layer Status	YES	Reason of enable/disable	3	"(invalid Base Station role)"	1	YES	
43	SIP name Resolution	NO	--	--	--	1	NO	6
44	Fail to Register to SIP	YES	Status Code (num) Server IP Address (textual)	500	"1 --> 10.54.6.8:5060"	1	NO	
45	Registration to SIP	YES	Server IP Address (textual)	--	"1 --> 10.54.6.8:5060"	1	YES	

Alarm Number	Alarm Label	Additional Data Present	Additional Data Type	Example (numeric)	Example (textual)	Default Consecutive Events to Toggle	User can set a desired amount for Consecutive Events	Note
46	Unregistration from SIP	NO	--	--	--	1	YES	7
47	SIP Server Change	NO	--	--	--	1	YES	6
48	SIP TRUNK Server	YES	Internal Data of the Trunk Server (textual)	--	"RA_SIP 15247"	1	NO	

Notes:

- 1: this is a common event for the two main supply voltages (12 V main supply and 3.3 V logic supply), so unless they are split into two distinct events, no additional data will be sent; currently, a more in-depth diagnostic can be obtained by **KAIROS_Manager**.
- 2: the value of the monitored parameters will be sent as additional data in a future release of the **ka_bst** software.
- 3: this is a common event for all the tuning voltages of the **PLLs** controlling the main oscillators and the TRX oscillators, and very often it raises together with an *unlock* event; to better identify the problem, a more in-depth diagnostic can be obtained by **KAIROS_Manager**.
- 4: this is a common event for all the *unlock* events of the **PLLs** controlling the main oscillators and the TRX oscillators, and sometimes it raises together with a *vtune* event; to better identify the problem, a more in-depth diagnostic can be obtained by **KAIROS_Manager**.
- 5: this event is not implemented currently.
- 6: this events are related to the use of **node names** instead of IP Addresses; the use of the names is not currently supported on **KAIROS**, so these events are not fully implemented.
- 7: this event is not implemented currently and will be fully operational in a future release of the **ka_bst** software.

SNMP EXTENSION

This method to dispatch Alarm or Notice Events is provided to send an SNMP Trap once an Event changes its status. To activate it you must configure at least one SNMP Server in the **Alarms Treatment** window and activate it on one or more event in the **Events Setup** window.

When an event occurs, the Base Station generates an SNMP Trap and sends it to every server defined as active for that event. The servers involved in this management, to properly handle the Trap, should have previously loaded the specific **MIB** file, distributed with the release of the software (see below, **SNMP MIB Files**).

The Trap is composed as an **SNMPv1-Trap**, according to **RFC 1157**, framed into an UDP packet:

IP Header	UDP Header	Version	Community	PDU Type	Enterprise	Agent Address	Generic Trap	Specific Trap	Up-Time	Variable Bindings
-----------	------------	---------	-----------	----------	------------	---------------	--------------	---------------	---------	-------------------

Below the meaning of the relevant fields:

Version:fixed value (**0**, that is SNMP version **1**),

Community:the value defined in the **Alarms Treatment** window,

PDU Type:fixed value (**4**, that is **Trap**),

Enterprise:enterprise code, composed of a **IANA** prefix (**1.3.6.1.4.1**) and the **IANA**-assigned **Personal Enterprise Number**; the **PEN** assigned to **Radio Activity** has the value **37755**, so the Enterprise value will be **1.3.6.1.4.1.37755**

Agent Address:the IP Address of the sender,

Generic Trap:code for the trap; currently is set to **6** (enterprise specific),

Specific Trap:Event Number as shown in the above table,

Up-Time:the system up-time: this is the amount of time in hundredths of seconds since the activation of the Alarms Events Subsystem.

The **Variable Bindings** field contains the data of the Trap, and is subsequently divided into several **OIDs**, each of which is composed as follows:

Enterprise	Traps ID	Value ID	Trailed Value
------------	----------	----------	---------------

Where:

Enterprise:see above for the *Enterprise Code*,

Traps ID:a fixed value identifying the **Radio Activity** product; for **KAIROS** Base Stations the value is **11**,

Value ID:a fixed value identifying the kind of the trailed value,

Trailed Value:the content of the trailed value.

The following table shows the contents of the **Value ID** and **Trailed Value** fields:

Value ID	SNMP MIB Label	SNMP Variable Type	Kind of Trailed Value	Description
1	ka-uptimer	INTEGER	Timestamp	The Linux timestamp (a 32-bits value representing the amount of seconds since 1970-01-01T00:00:00) of the instant of the event occurrence
2	ka-eqp-id	INTEGER	EQP UID	The <i>Unique ID</i> of the Base Station (unique for each Radio Activity Equipment, automatically assigned)
3	ka-eqp-sn	OCTET STRING	EQP S/N	The Base Station Serial Number, assigned by Radio Activity during the production process and reported in the main window and in the Equipment Info window of KAIROS_Manager (in the format <BBB KA NNNN>, where BBB is a 3-digits frequency band code, KA is a fixed code that identify the KAIROS product and NNNN is a 4-digits progressive number)
4	ka-eqp-name	OCTET STRING	EQP NAME	The User assigned name of the Base Station, set in KAIROS→Configuration→Main Setup)
5	ka-alarm-id	INTEGER	Alarm Number	The Event Internal Number, the same as shown in the above table, but starting from 0 instead than 1
6	ka-item-id	INTEGER	Device	The internal device where the Event was generated (currently not used, forced to 0)
7	ka-alarm-status	INTEGER (0..1)	Event Status	Its value is 1 for a RAISE Event or a NOTICE Event , and 0 for a CLEAR Event
8	ka-alarm-severity	INTEGER (0..4)	Severity	The value(s) of the severity degree set in the Events Setup window, in the User/System-Defined Severity listbox of the Alarm Raise section and in the User-Defined Severity / Common Severity Degree listbox of the Alarm Clear section (as shown above, in Severity Degree of an Alarm Event and Events Setup),
9	ka-user-level	INTEGER (0..15)	User-Defined Value	Its content is the value(s) set in the User-Defined Level textboxes of the Alarm Raise and Alarm Clear sections of the Events Setup window, distinct for a RAISE Event and a CLEAR Event
10	ka-aux-value	INTEGER	Additional Data (numeric)	Additional data (numeric), which can contain the value of an analog parameter or a code value
11	ka-aux-text	INTEGER	Additional Data (textual)	Additional data (textual), which can contain the value of an analog parameter or a code value. This value can differ from the <i>numeric additional data</i> and can also be merged into the related text (see Events Setup)
12	ka-event-text	OCTET STRING	Related Text	Event Related Text

So, for every Trap a **KAIROS** will send this series of UDP packets to the SNMP Server (you see in *italic* the fixed values):

Packet #1 →	IP Header	UDP Header	0	Community	4	1.3.6.1.4.1.37755	Agent Address	6	Event ID	Up-Time	1.3.6.1.4.1.37755	11	1	Timestamp
Packet #2 →	IP Header	UDP Header	0	Community	4	1.3.6.1.4.1.37755	Agent Address	6	Event ID	Up-Time	1.3.6.1.4.1.37755	11	2	EQP UID
Packet #3 →	IP Header	UDP Header	0	Community	4	1.3.6.1.4.1.37755	Agent Address	6	Event ID	Up-Time	1.3.6.1.4.1.37755	11	3	EQP S/N
Packet #4 →	IP Header	UDP Header	0	Community	4	1.3.6.1.4.1.37755	Agent Address	6	Event ID	Up-Time	1.3.6.1.4.1.37755	11	4	EQP NAME
Packet #5 →	IP Header	UDP Header	0	Community	4	1.3.6.1.4.1.37755	Agent Address	6	Event ID	Up-Time	1.3.6.1.4.1.37755	11	5	Alarm Number
Packet #6 →	IP Header	UDP Header	0	Community	4	1.3.6.1.4.1.37755	Agent Address	6	Event ID	Up-Time	1.3.6.1.4.1.37755	11	6	0
Packet #7 →	IP Header	UDP Header	0	Community	4	1.3.6.1.4.1.37755	Agent Address	6	Event ID	Up-Time	1.3.6.1.4.1.37755	11	7	Event Status
Packet #8 →	IP Header	UDP Header	0	Community	4	1.3.6.1.4.1.37755	Agent Address	6	Event ID	Up-Time	1.3.6.1.4.1.37755	11	8	Severity
Packet #9 →	IP Header	UDP Header	0	Community	4	1.3.6.1.4.1.37755	Agent Address	6	Event ID	Up-Time	1.3.6.1.4.1.37755	11	9	User-Defined Value
Packet #10 →	IP Header	UDP Header	0	Community	4	1.3.6.1.4.1.37755	Agent Address	6	Event ID	Up-Time	1.3.6.1.4.1.37755	11	10	Additional Data (numeric)
Packet #11 →	IP Header	UDP Header	0	Community	4	1.3.6.1.4.1.37755	Agent Address	6	Event ID	Up-Time	1.3.6.1.4.1.37755	11	11	Additional Data (textual)
Packet #12 →	IP Header	UDP Header	0	Community	4	1.3.6.1.4.1.37755	Agent Address	6	Event ID	Up-Time	1.3.6.1.4.1.37755	11	12	Related Text

ADDITIONAL DATA FIELDS ON SNMP TRAPS

The **Additional Data** in the **SNMP Extension** are carried in two distinct fields: **ka-aux-value** (numeric) and **ka-aux-text** (textual). In addition, the User can decide to add the **Additional Data** at the end of the **Related Text** (see **<SNMP Traps / Traps Sending Policy> Section**)

While the **Additional Data** are present only for some Alarm Events (as we have already seen in the **Additional Data** paragraph), the related **OIDs** in the Trap are always present; when no **Additional Data** are inserted, the numeric field value is set to **0**, the textual field value is set to **"N/A"**, and no content is appended to the event-related text message, even if the user has set this option.

HANDLING THE SEVERITY DEGREE

While for some Alarms the Severity Degree for **RAISE Event** is internally assigned and cannot be changed, for some other Alarms the User can assign a desired Severity Degree, depending on the context in which the Base Station is placed. The User can however choose a desired Severity Degree for every **CLEAR Event** - or a common degree if he set this option (see above, **<SNMP Traps / SNMP Servers> Section**) – if the **CLEAR Event** is provided for the specified Event.

The following table shows the Severity Degrees assigned by default and the capabilities of the User:

<i>Alarm Number</i>	<i>Alarm Label</i>	<i>Default Degree for RAISE Event</i>	<i>CLEAR Event provided</i>	<i>Default Degree for CLEAR Event</i>	<i>User can set a desired Degree for RAISE Event</i>
1	I/O 1 Contact	WARNING	YES	WARNING	YES
2	I/O 2 Contact	WARNING	YES	WARNING	YES
3	Logic Supply Status	CRITICAL	YES	CRITICAL	NO
4	Supply Undervoltage	MAJOR	YES	MAJOR	NO
5	Supply Overvoltage	MAJOR	YES	MAJOR	NO
6	Ethernet Link Status	MAJOR	YES	MAJOR	NO
7	PLD Data Correctness	CRITICAL	YES	CRITICAL	NO
8	DSP Loading/Starting	CRITICAL	YES	CRITICAL	NO
9	GNSS Status	MAJOR	YES	MAJOR	NO
10	Vocoders Status	MAJOR	YES	MAJOR	NO
11	BS Temperature	MAJOR	YES	MAJOR	NO
12	TX Temperature	MAJOR	YES	MAJOR	NO
13	No TX Power	CRITICAL	YES	CRITICAL	NO
14	TX Power too low	MAJOR	YES	MAJOR	NO
15	TX Power too high	MAJOR	YES	MAJOR	NO
16	TX SWR Warning	MAJOR	YES	MAJOR	NO
17	TX SWR Alarm	CRITICAL	YES	CRITICAL	NO
18	TX Power Reduction	CRITICAL	YES	CRITICAL	NO
19	Synchronization Source	WARNING	YES	WARNING	YES
20	Synchronization Status	WARNING	YES	WARNING	NO
21	Board Vtunes Status	CRITICAL	YES	CRITICAL	NO
22	TRX Vtunes Status	CRITICAL	YES	CRITICAL	NO
23	Board Clocks Status	CRITICAL	YES	CRITICAL	NO
24	TRX Clocks Status	CRITICAL	YES	CRITICAL	NO
25	Board PLLs Lock Status	CRITICAL	YES	CRITICAL	NO
26	TRX PLLs Lock Status	CRITICAL	YES	CRITICAL	NO
27	PLD Status	CRITICAL	YES	CRITICAL	NO
28	PLD<->DSP Chan Status	CRITICAL	YES	CRITICAL	NO
29	RX IFs Status	CRITICAL	YES	CRITICAL	NO
30	RXs Status	CRITICAL	YES	CRITICAL	NO
31	RF Channel Noise	MINOR	YES	MINOR	YES
32	Registration to Master	MAJOR	YES	MAJOR	YES

<i>Alarm Number</i>	<i>Alarm Label</i>	<i>Default Degree for RAISE Event</i>	<i>CLEAR Event provided</i>	<i>Default Degree for CLEAR Event</i>	<i>User can set a desired Degree for RAISE Event</i>
33	Registration of Slave	NOTICE	NO	NOTICE	YES
34	Unregistration of Slave	NOTICE	NO	NOTICE	YES
35	Loss of Slave	MAJOR	NO	MAJOR	YES
36	Master Role	WARNING	YES	WARNING	YES
37	Connection to BK MST	WARNING	YES	WARNING	YES
38	DMR Emergency Call	WARNING	NO	WARNING	YES
39	1+1 ACTIVE State	NOTICE	NO	NOTICE	YES
40	1+1 HOT-SPARE State	NOTICE	NO	NOTICE	YES
41	TRX Layer Status	WARNING	YES	WARNING	YES
42	BS Layer Status	WARNING	YES	WARNING	YES
43	SIP name Resolution	MAJOR	YES	MAJOR	NO
44	Fail to Register to SIP	NOTICE	NO	NOTICE	NO
45	Registration to SIP	NOTICE	NO	NOTICE	YES
46	Unregistration from SIP	NOTICE	NO	NOTICE	YES
47	SIP Server Change	NOTICE	NO	NOTICE	YES
48	SIP TRUNK Server	MAJOR	YES	MAJOR	NO

SNMP MIB FILES

To properly handle the SNMP Traps (i.e.: report an anomaly, log events per-item, apply filters, interact with databases, build statistics...), all the servers involved in the management should have previously loaded into their SNMP Management System the **MIB** file for **KAIROS** (*KAIROS-SNMP-TRAPS.mib*), released along with **KAIROS_Manager**. This file specifies how to handle the trap data and contains the rules according to which the trap data are formatted.

The **MIB** file for **KAIROS** has several “dependencies”, which means that some other MIB files must be already present or loaded in your SNMP management system to handle the **KAIROS MIB** file correctly. These needed MIB files are anyway present in the installation package of **KAIROS_Manager**.

You should upload on your SNMP management system (if you haven’t already) and possibly compile the set of MIB files in this order:

- ∞ rfc1158.mib,
- ∞ RFC1155.mib,
- ∞ RFC-1215.mib,
- ∞ RFC1213.mib,
- ∞ RFC-1212.mib,
- ∞ KAIROS-SNMP-TRAPS.mib.

ON-AIR MESSAGES EXTENSION

This method to dispatch Alarm or Notice Events is provided to send one or more DMR Text Messages once an Event change its status, up to 4 DMR Destinations, group or individual. To activate it you must configure at least one **DMR Destination** in the **Alarms Treatment** window and activate it on one or more event in the **Events Setup** window.

You can then set how many times the message will be sent (repetitions). If you set more than one time, you can also set an interval between consecutive times.

Its operating mode is very similar to what described for the **SNMP Extension**, with these differences:

- ∞ the **Additional Data** are always merged into the DMR Message, as trailing data of the **Event Related Text**,
- ∞ the DMR Messages are sent regardless of the Severity Degree of the Event,
- ∞ the **Alarm Events Subsystem** will send one or more DMR Message for a **RAISE Event** and one or more DMR Message for a **CLEAR Event**, that is, setting the repetition of the **RAISE Alarm** instead of sending the **CLEAR Alarm** has no effect on this Extension,
- ∞ the User can optionally set a premise to the Message body, composed by date, time and an identifier of the Base Station.

For the related parameters and settings, please review **<DMR Messages / Destination> Section** and **<DMR Messages / Message Parameters> Section**.